A Deep Transfer Operator Learning Method for Temperature Field Reconstruction in a Lithium-Ion Battery Pack

Yuchen Wang, *Student Member, IEEE*, Can Xiong, Changjiang Ju, Genke Yang, Yu-wang Chen, and Xiaotian Yu

***Abstract*—Nonuniform thermal behavior in lithium-ion battery packs can accelerate aging, leading to inconsistent cell performance. If not adequately monitored and managed, this heating can give rise to unwanted side reactions, fires, and explosions, underscoring the criticality of temperature field reconstruction. In recent years, data-driven methods have gained popularity for addressing the temperature field reconstruction problem. However, many existing data-driven approaches require retraining when system parameters change, such as the initial temperature distribution or working conditions. This article presents a deep transfer operator learning method named physics-informed adversarial networks. The model architecture incorporates transformer blocks to capture comprehensive time and space features. Additionally, to enhance interpretability and generalization, the model introduces two effective mechanisms: 1) the integration of thermal partial differential equations to ensure compliance with physical laws; and 2) the application of domain adversarial mechanism in transfer learning to extract domain-invariant feature representations. These mechanisms enable the model to effectively reconstruct the temperature field, even in unencountered scenarios during training. The proposed method is validated under real-world energy storage working conditions, demonstrating superior performance compared to state-of-the-art deep learning methods. Notably, the approach exhibits excellent performance even when confronted with the limited availability of training data.**

***Index Terms*—Deep operator learning, lithium-ion battery (LiB) pack, temperature field reconstruction (TFR), transfer learning, transformer.**

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Yuchen Wang, Can Xiong, Changjiang Ju, and Genke Yang are with the Ningbo Artificial Intelligence Institute, Department of Automation, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: wyc_077@sjtu.edu.cn; candemailbox@sjtu.edu.cn; juchangjiang@sjtu.edu.cn; gkyang@sjtu.edu.cn).

Yu-wang Chen is with the Manchester Business School, The University of Manchester, M15 6PB Manchester, U.K. (e-mail: yu-wang.chen@manchester.ac.uk).

Xiaotian Yu is with the Ningbo Artificial Intelligence Institute, Ningbo 315000, China (e-mail: yuxiaotian@sjtu-naii.com).

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I. Introduction

LITHIUM-ION batteries (LiBs) have become the best choice for temporal electrochemical energy storage and the electric vehicle field since they have the characteristics of long cycle life and high power density [1]. At the same time, it is considered thermally fragile. Enormous heat is generated during charging and discharging. Without prompt cooling, the battery temperature will rise rapidly, triggering a series of side reactions, and eventually leading to thermal runaway [2]. Hence, monitoring the battery temperature in real time is vital to system safety and has attracted increasing research attention.

The thermal model of lithium batteries plays a crucial role in comprehending and forecasting temperature variations. Numerous studies have extensively delved into the thermal modeling of the single LiB cell [3], [4], [5], [6], [7]. Notably, in a study by Yao et al. [6], the temperature state of the battery was treated as a time series, and an RNN-based model was employed to predict the temperature at the next time step. In such instances, time series pattern recognition models [8], [9], [10] that address data uncertainty can be considered to enhance prediction accuracy. However, for practical applications necessitating higher power output, it is customary to assemble batteries into packs. The installation of temperature sensors for each battery within the pack becomes impractical. As a result, cell-level techniques become infeasible due to increased dimensions and the complex interplay of multiple heat sources [11]. Furthermore, temperature nonuniformity commonly arises within battery packs, hastening aging and exacerbating cell inconsistency [12]. Hence, real-world applications prioritize the assessment of temperature field distribution. Reconstructing the temperature field of battery packs can significantly enhance the efficacy of thermal management control methods [13], providing multidimensional thermal information beneficial for thermal runaway forecasting [2].

The LiB pack is a typical distributed parameter system, and the nonuniform thermal behavior within it can be described by partial differential equations (PDEs). Several researchers are dedicated to investigating the temperature field reconstruction (TFR) problem using model-based approaches. Wang et al. [14] employed the computational fluid dynamics method to deduce the temperature field distribution by solving heat PDEs. Despite the precise information obtained through numerical approaches

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regarding cell temperature gradients, they prove unsuitable for real-time temperature estimation due to significant computational costs.

To simplify the analysis, [11] and [15] converted the heat PDEs into ordinary differential equations (ODEs) and developed a state observer to determine the temperature status of nodes. The above state observer-based methods often formulate the state space model to generate a linear time-invariant form for analysis under specific assumptions. As a result, the performance inevitably declines as the accuracy of the system model decreases.

Model-based approaches require comprehensive knowledge of the physical parameters of the system. In practice, some parameters of the system are difficult to obtain [16]. To address this issue, recent studies focus on modeling the battery thermal process by extracting patterns and correlations from extensive data. Many of these studies adopt a time/space (T/S) separation framework to reconstruct the temperature field. The authors in [16] and [17] applied Karhunen–Loève decomposition-based methods to learning spatial basis functions (SBFs) and used neural networks to approximate the temporal dynamics. Nevertheless, these methods suffer from a significant drawback: any alteration in the initial or boundary conditions of the system necessitates retraining the spatio-temporal model. The retraining strategy requires high computational cost and is not suitable for real-time applications. Thus, Zhou et al. [18] proposed an incremental training approach to effectively update SBFs and temporal model parameters. Nonetheless, this approach still requires multiple sensors to recollect data, such as 20 sensors for one prismatic lithium-ion cell, which is impractical in real-world applications. It is worth noting that the aforementioned studies solely investigate the thermal characteristics of a single prismatic lithium-ion cell in a controlled lab environment, while the thermal behavior within a battery pack and real-world applications presents even greater challenges.

In this study, our goal is to develop a TFR model using few temperature sensors as inputs and capable of effectively generalizing across diverse boundary and initial conditions without the need for retraining. This necessitates the model to learn a mapping (defined as "operator") from monitoring sensor points to the temperature field. Standard neural networks can be employed to approximate this operator. Among them, Deep operator learning methods [19], [20], which significantly reduce the generalization error compared to fully connected networks when learning operators in PDE problems [20], are first introduced in TFR problem. Our method is based on one of the most advanced and widely known operator learning frameworks called deep operator networks (DeepONet) [20]. The time-dependent and space-dependent features are decoupled and used as inputs. As the multilayer perceptron (MLP) structure of vanilla DeepONet cannot fully leverage temporal and spatial information, we novelty introduce the architecture of Transformer [21], [22] to reconstruct the temperature field more efficiently. Additionally, the performance of most deep learning methods tends to decline on unseen data due to distributional differences between the training and testing datasets. In industrial scenarios, the performance of methods on unseen data is critical because it is challenging to capture all potential working conditions within the collected data, given the dynamic nature of the environment. To address this challenge, we utilize domain adaptation methods in transfer learning to enhance the performance of deep operator learning methods on out-of-domain cases.

In summary, our objective is to explore a data-driven method that achieves improved generalization and interpretability for reconstructing the temperature field in LiB packs. Building upon the DeepONet framework, we first propose a novel approach named physics-informed adversarial networks (PIAN) for this purpose. The main contributions of this work are outlined below.

1) PIAN is proposed to reconstruct the temperature field in a battery pack, capturing its nonuniform thermal characteristics and exhibiting strong generalization under various boundary and initial conditions without retraining.
2) The heat PDEs are embedded to the model during training, making the output of PIAN conform to the laws of physics.
3) Domain-invariant features are extracted from PIAN using transfer learning techniques, enabling enhanced performance on out-of-domain data.
4) The method is validated under various energy storage working conditions and compared with state-of-the-art deep learning methods. Experimental results demonstrate its significant superiority in terms of prediction accuracy and out-of-domain generalization.

The rest of this article is organized as follows. Section II examines the study subject and models it using the PDE equation. The learning objectives for reconstructing the temperature field are established, and the inputs and outputs are determined in this section. In Section III, we present the proposal method. Section IV provides an overview of the experimental setup and implementation details. Next, Section V presents the results of experiments. Finally, Section VI concludes this article.

II. Related Work

In this section, we present the relevant research findings on deep operator learning and discuss recently proposed methods for operator learning. Additionally, we review the literature on domain adaptation methods in transfer learning, which focuses on creating and establishing models that can be generalized in different domains.

A. Deep Operator Learning

The concept of deep operator learning was first introduced through the development of DeepONet [20]. This approach aims to learn the mapping between function spaces, enabling trained neural networks to accurately predict the system's response in real-time for various boundary and initial conditions without the need of additional extensive training. As a result, deep operator learning methods can effectively replace traditional expensive ODE/PDE solvers, providing more efficient solutions. Since the introduction of DeepONet, several other deep operator learning methods have been proposed. For instance, the graph kernel network (GKN) operator learning for solving PDEs is presented in [23]. Additionally, an alternative architecture called

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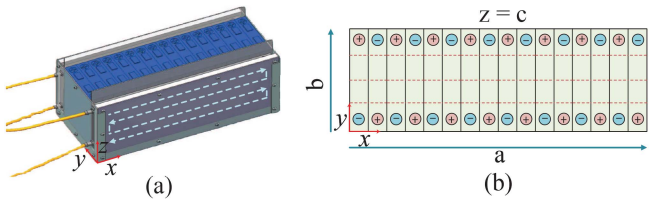


Fig. 1. (a) Schematic diagram of the physical structure of the battery pack. (b) Electrode surface in a battery pack studied in this work.

the Fourier neural operator (FNO) is introduced in [24]. The FNO formulates operator regression by directly parameterizing the integral kernel in Fourier space. The authors in [25] and [26] demonstrated that GKN and FNO were distinct realizations of DeepONet, with appropriate adjustments made to the architecture. Subsequently, variants and extensions [27], [28], [29] have been developed based on these three frameworks: DeepONet, GKN, and FNO. Several studies have successfully applied deep operator learning methods in fluid mechanics [19] and biological fields [30], yielding promising results. Therefore, the introduction of deep operator learning holds significant importance for modeling and state estimation of complex industrial distributed parameter systems.

B. Domain Adaptation

Deep learning methods typically assume that the training and testing sets are independent and identically distributed. If there are differences in the data distribution between the training and testing sets, the problem of covariate shift may arise, leading to a decrease in the model's performance on the testing set. To address this issue, domain adaptation methods have been proposed to reduce the distribution difference between the training set and the testing set, thereby improving the model's generalization ability. Recent domain adaptation methods minimize the distribution shift between training and testing domains [31] mainly through five approaches: statistics matching, domain adversarial learning, hypothesis adversarial learning, domain translation, and semisupervised learning. The first three methods are commonly used and are derived from $H\Delta H-$divergence [32], [33] and disparity discrepancy [34] theories. Methods for statistics matching is based on nonparametric distances, which are expressive but data-efficient, making them suitable for small data applications. On the other hand, domain adversarial learning and hypothesis adversarial learning rely on parametric distances, which can be evaluated during learning, therefore perform better with increasing data volumes. Domain adaptation can improve the performance of algorithms on unseen data, which is one of the most important aspect in industrial applications.

III. Problem Set up

This study focuses on an energy storage system comprising multiple parallel-connected LiB packs. Each battery pack consists of 16 prismatic LiFePO4 (LFP) lithium-ion cells connected in series and enclosed within a metal casing. Fig. 1(a) illustrates the 3-D structure of a battery pack with dimensions of $a \times b \times c$. There are liquid cooling panels on both sides of the battery pack. To dissipate heat, water flowing into the channel is controlled by two pumps. According to Mevawallaet al. [35], the surface temperature near the prismatic LFP cell poles was often the highest. In other words, thermal inhomogeneity is more likely to appear on the electrode surface in a battery pack since generated heat conducts the most there. Moreover, it is less recommended to mount sensors in other locations than the electrode surface, due to the compact design of battery packs and cost saving in manufacturing. Therefore, our objective is to reconstruct the temperature distribution at the electrode surface, as illustrated in Fig. 1(b).

The temperature distribution $T(t, x, y, z)$ can be obtained by solving a simplified 3-D PDE based on the thermal energy conservation principle [11]

$$\rho C_p \frac{\partial T}{\partial t} = \lambda_x \frac{\partial^2 T}{\partial x^2} + \lambda_y \frac{\partial^2 T}{\partial y^2} + \lambda_z \frac{\partial^2 T}{\partial z^2} + Q_{\text{heat}}. \quad (1)$$

As shown in (1), where t, ρ, C_p, λ are the experiment time, average density, heat capacity, and thermal conductivity along x, y, z axis, respectively. Variable Q_{heat} represents the heat generated by the battery pack during operation as given in the following:

$$Q_{\text{heat}} = \frac{a(x, y, z)I}{v}\left[U_{\text{ocv}} - U_t - T\frac{dU_{\text{ocv}}}{dT}\right]. \quad (2)$$

The open circuit voltage (OCV) of the battery, denoted as U_{ocv}, is a function of the state of charge (SOC). The variables I and U_t correspond to the current (positive for discharge, negative for charge) and terminal voltage, respectively. The symbol v represents the volume of the battery pack. $\frac{dU_{\text{ocv}}}{dT}$ is the coefficient of entropic heat generation, and the function $a(x, y, z)$ captures the spatial variations of heat generation along axis.

In fact, the temperature distribution is also dependent on electrical and environmental parameters. The physical variables considered include the current (I), voltage (U_t), SOC of the battery, and the temperatures of the ambient air (T_{amb}) and cooling liquid (T_{liquid}). We define the physical parameter space as $\mathcal{L}$, and the geometric parameters space as $\mathcal{X}$, both of which are bounded as shown in the following:

$$\begin{aligned}\mathcal{X} &= \left\{(x, y, z)^\tau \in \mathbb{R}^3; 0 \le x \le a, 0 \le y \le b, 0 \le z \le c\right\}\\ \mathcal{L} &= \Big\{(U_t, I, \text{SOC}, T_{\text{amb}}, T_{\text{liquid}})^\tau \in \mathbb{R}^5\\ &\quad U_{t_{\min}} \le U_t \le U_{t_{\max}}, -I_{\max} \le I \le I_{\max}\\ &\quad 0 \le \text{SOC} \le 1, T_{\text{amb}_{\min}} \le T_{\text{amb}} \le T_{\text{amb}_{\max}}\\ &\quad T_{\text{liquid}_{\min}} \le T_{\text{liquid}} \le T_{\text{liquid}_{\max}}\Big\} \end{aligned} \quad (3)$$

where the subscripts "min" and "max" indicate the minimal and maximal values chosen in the experiment. The elements of corresponding spaces are denoted as $X \triangleq (x, y, z)^\tau \in \mathcal{X}$ and $L \triangleq (U_t, I, \text{SOC}, T_{\text{amb}}, T_{\text{liquid}})^\tau \in \mathcal{L}$. To facilitate description, the variables, sampled values, and predicted values are represented by L, $\bar{L}$, and $\widehat{L}$, respectively. The same notation applies to other variables mentioned in the subsequent paragraphs.

Let's assume that the input space $\mathcal{D}$ is a product space of $\mathcal{X}$ and $\mathcal{L}$, denoted as $\mathcal{D} = \mathcal{X} \times \mathcal{L}$. Each element in $\mathcal{D}$ is represented as d, where $d \in \mathcal{D}$, $d_i(t) \triangleq [X_i; L(t, X_i)]$, $i = 1, 2, \ldots, N$. N

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represents the total validation location in the geometric domain. The range of temperature is also bounded as follows (4). The upper bound value $T_{\max}$ is empirically determined. This decision is based on the maximum allowable operating temperature of the LiB

$$\mathcal{V} = [0, T_{\max}]. \tag{4}$$

It is assumed that the temperature of the contact surface between cells is continuous, since adjacent cells are placed closely. The cooling and near-adiabatic effects are included in the boundaries of the battery pack's thermal model. The boundary conditions are listed as follows:

$$\begin{aligned}
\lambda_x \frac{\partial T}{\partial x}|_{x=0} &= h_{\text{conv1}}(T - T_{\text{amb}}) \\
-\lambda_x \frac{\partial T}{\partial x}|_{x=a} &= h_{\text{conv1}}(T - T_{\text{amb}}) \\
\lambda_y \frac{\partial T}{\partial y}|_{y=0} &= h_{\text{conv2}}(T - T_{\text{liquid}}) \\
-\lambda_y \frac{\partial T}{\partial y}|_{y=b} &= h_{\text{conv2}}(T - T_{\text{liquid}}) \\
\lambda_z \frac{\partial T}{\partial z}|_{z=0} &= 0 \\
-\lambda_z \frac{\partial T}{\partial z}|_{z=c} &= h_{\text{conv3}}(T - T_{\text{amb}}) \\
T|_{\text{cell}_r} &= T|_{\text{cell}_{r+1}}, r = 1, \ldots, 15
\end{aligned} \tag{5}$$

where h_{conv1}, h_{conv2}, and h_{conv3} denote the corresponding heat transfer coefficient along the boundaries.

Traditionally, as proposed in [16], it is is to try to learn a mapping g to predict the temperature (6), which is asymptotic of $T(t, X, L)$

$$g : [0, t_{\max}] \times \mathcal{D} \to \mathcal{V}. \tag{6}$$

This approach assumes that the system is under a specific initial condition, for instance, the initial temperature is identical and equal to a certain value T_0 as follows:

$$g\left(0, X_i, \bar{L}\right) = T_0, i = 1, 2, \ldots, N. \tag{7}$$

Furthermore, within the entire time interval $[0, t_{\max}]$, we discrete the interval and assumed to predict the temperature field at M times. Consequently, the TFR task can be transformed to an optimization problem as follows:

$$\begin{aligned}
&\min_{g} \left(\sum_{j=1}^{M} \sum_{i=1}^{N} \left| g\left(t_j, d_i\left(t_j\right)\right) - \bar{T}\left(t_j, d_i\left(t_j\right)\right) \right| \right) \\
&s.t.\ (1) - (7)
\end{aligned} \tag{8}$$

where $\bar{T}(t_j, d_i(t_j))(i = 1, 2, \ldots, N)$ denotes the temperature value obtained by the ith monitoring sensors at $(t_j, d_i(t_j))$.

This kind of method essentially approximates the solution of a PDE given specific initial and boundary conditions. Consequently, when facing new data under different initial conditions, for instance, $g(0, X, L) = h(X, L)$, where h is an unknown function that is different from (7). This method may necessitate gathering data across diverse initial conditions for retraining to achieve accurate temperature predictions.

To address this issue, it is preferred to directly learn a mapping that represents the solutions of PDEs for varying initial and boundary conditions within an appropriate range. In our approach, a subset of p sensors from N available positions is selected and numbered $[1] - [p]$, where $p \ll N$. These sparse temperature values obtained from the p sensors can be considered as the specified initial conditions of the system, for which we can represent it by $u^{(j)}$

$$\begin{aligned}
u^{(j)} &\triangleq \left[\bar{T}\left(t_j, d_1\left(t_j\right)\right), \bar{T}\left(t_j, d_2\left(t_j\right)\right), \ldots, \bar{T}\left(t_j, d_p\left(t_j\right)\right)\right] \\
&j = 1, \ldots, M.
\end{aligned} \tag{9}$$

We aim to obtain the temperature field solutions $T(t, X, L) \in C^2[[0, t_{\max}] \times \mathcal{D}, \mathcal{V}]$ under different initial conditions $u^{(j)}$. We assume the existence of a unique solution T for any $u^{(j)}$, which allows us to define the mapping as an operator $\mathcal{G}$. However, due to the nature of network approximations, discretization is necessary even though our objective is to learn an operator in function spaces. In this study, we discretize the surface uniformly into N grids. Considering the time-delay characteristics of the physical system, we incorporate information from the previous s time windows. Therefore, the definition of $\mathcal{G}$ is provided in the follows:

$$\mathcal{G} : \mathbb{R}^p \times [[0, t_{\max}] \times \mathcal{D}]^s \to \mathbb{R}^N. \tag{10}$$

In this study, we specifically set $p = 8$, $N = 64$, and $s = 10$, with each adjacent sampling interval being 1 min. Furthermore, we extend the variable d by incorporating the time coordinate t, which enables the incorporation of physics-informed mechanisms discussed in detail in Section IV. We define $d_i^{(j)} = [d_i(t_{j-s+1}), \ldots, d_i(t_{j-1}), d_i(t_j)]^\tau$. The optimization objective is finally rewritten as follows:

$$\begin{aligned}
&\min_{\mathcal{G}} \left(\sum_{j=s}^{M+s-1} \sum_{i=1}^{N} \left| \mathcal{G}\left(u^{(j)}\right)\left(d_i^{(j)}\right) - \bar{T}(t_j, d_i(t_j)) \right| \right) \\
&s.t.\ (1) - (5).
\end{aligned} \tag{11}$$

Comparing (11) to (8), the initial condition (7) is no longer treated as the constraints, nonetheless, the influence of the initial condition to the output is learned by our specially designed optimization objective, enabling better feasibility. It is proved by the universal approximation theorem [36] that deep neural networks are capable of approximating the operator $\mathcal{G}$.

IV. Method

Our method aims to learn the operator $\mathcal{G}$. A critical consideration in this process is the determination of u, which, in this study, is associated with sensor placement. We utilize the collected training data to determine the placement of the sensors. Once u is determined, we can approximate $\mathcal{G}$ under the DeepONet framework. It is noteworthy that the selection of sensor locations is prior. During the testing stage, the algorithm will accept real-time temperature values collected by fixed sensors, along

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with information such as voltage, current, ambient temperature, and water temperature, to reconstruct the temperature field.

A. Sensor Placement Strategy

It is crucial for the input function u to preserve high-fidelity spatial information based on sensor placement. In this work, we adopt a data-driven strategy [37] to get a feasible solution for the sensor placement in a LiB pack. Consider a substantial amount of temperature data, denoted by W, that has been sampled on the 2-D surface of a lithium battery pack. The snapshot of the temperature field across M times is represented by $w_j \in \mathbb{R}^N$, as shown in the following:

$$W = [w_1, w_2, \ldots, w_M] \in \mathbb{R}^{N\times M} \tag{12}$$

where N is referred to as the discrete values of the temperature field, specifically 64 in this study. We want to reduce this high-dimensional data W. Thus, a proper orthogonal decomposition [38] (POD) method is performed on the state W, for the each element w_j can be represent by the following:

$$w_j \approx \Psi_r a(t_j) \quad j = 1, \ldots, M \tag{13}$$

where $a(t_j) = [a_1(t_j), a_2(t_j), \ldots, a_r(t_j)]^\tau$ is a time-varying coefficient, the eigenmodes Ψ_r are only spatially dependent and do not vary with time and can be obtained by truncated singular value decomposition (truncated SVD) of W, as shown in (14). r is the rank value in truncated SVD

$$W = \Psi \Sigma V^{\mathrm{T}} \approx \Psi_r \Sigma_r V_r^{\mathrm{T}}. \tag{14}$$

Thus, the high-dimensional data W can be represented by POD coefficients a, where $a = [a(t_1), a(t_2), \ldots, a(t_M)]$, which theoretically correlated to sensor observations l

$$\begin{aligned} l &= [l_1, l_2, \ldots, l_M] \in \mathbb{R}^{p\times M} \\ l_j &= C w_j = C\Psi_r a\left(t_j\right) \end{aligned} \tag{15}$$

where p is the number of selected sensors and $C \in \mathbb{R}^{p\times M}$ is the sensor measurement matrix defined as in the following:

$$C = \left[e_{\gamma_1}, e_{\gamma_2}, \ldots, e_{\gamma_p}\right]^\tau \tag{16}$$

e_{γ_p} represents the standard basis vector of $\mathbb{R}^N$, where the value is 1 at index γ_p and zero elsewhere. When w_i is unknown, reconstruction the temperature field $\widehat{w_j}$ can be done with unknown $\widehat{a(t_j)}$ based on Moore–Penrose pseudoinverse

$$\begin{aligned} \widehat{w_j} &= \Psi_r \widehat{a(t_j)} = \Psi_r \Theta^{-1} l_j \\ \Theta &= C\Psi_r. \end{aligned} \tag{17}$$

Thus, the combination of (15)–(17) reveals that optimal sensor placement results in maximum fidelity of the temperature field $\widehat{w_j}$. The sensor placement problem is transformed to find the rows of Ψ_r that correspond to the optimization matrix Θ condition inversion.

First, we should determine the sensor number p. In this article, it is important to note that oversampling, where $p > r$, is not considered for the LiB pack due to cost and safety concerns. Therefore, the number of sensors, p, is equal to r. Selecting the intrinsic target rank r without amplifying data noise is challenging. Some studies [39], [40] have established the optimal threshold for r based on the distribution of singular values and the aspect ratio of the data matrix, under the assumption of additive Gaussian white noise with unknown variance. Determining the true noise distribution of the sensors is challenging due to potential interference from environmental factors and the inherent noise of the measurement circuit, both of which can impact the final noise distribution. In this study, we adopt the approach proposed in [37] to determine the value of r by thresholding the singular value and reconstructing the temperature field error at varying r values using (17). Specifically, we identify the value of r based on the inflection point in the exponential decay rate of the singular values and the reconstruction error. At this point, the first r POD modes effectively capture the majority of the system's energy.

Then, we aim to determine the location of the sensors, which is equal to optimize the condition number of Θ. The condition number of Θ can be indirectly limited by optimizing its determinant, as demonstrated in the following:

$$\widehat{\gamma} = \underset{\gamma, |\gamma|=p}{\arg\max} \left|\det \Theta\right|. \tag{18}$$

This article utilizes a heuristic greedy sampling approach [37]. The column pivoting of Ψ_r^{T} using QR factorization as follows generates p optimally sampled sensors for r basis modes:

$$\psi_r^\tau C^\tau = QR. \tag{19}$$

This approach is effective because the pivoting process approximates a greedy solution for optimizing (18), also referred to as submatrix volume maximization, due to the matrix volume being equal to the absolute value of the determinant.

B. Physics-Informed Adversarial Networks

In this section, we propose a deep transfer operator learning methods to learn the operator $\mathcal{G}$. Vanilla DeepONet accepts coordinate d and input function u as inputs. It employs two subnetworks to process these inputs: the trunk network and the branch network. The trunk network takes coordinate d as input and yields a scalar output vector $[z_1, z_2, \ldots, z_h]^\tau \in \mathbb{R}^h$. On the other hand, the branch network accepts the input vector $[u(d_1), u(d_2), \ldots, u(d_p)]^\tau$ and produces another scalar vector $[b_1, b_2, \ldots, b_h]^\tau \in \mathbb{R}^h$. These two network outputs are then combined to generate $\mathcal{G}(u)(d)$, as illustrated in the following:

$$\mathcal{G}(u)(d) = \sum_{k=1}^{h} b_k z_k. \tag{20}$$

In this study, given the 3-D expression of the thermal model, we designate the positional coordinates of the sensors as 3-D points $X_i = (x_i, y_i, z_i)$. This designation aligns with (1) and facilitates the subsequent integration of the differential equation into the neural network training process. We introduce $d_i(t) \triangleq [X_i; t; L(t, X_i)]$, where i ranges from 1 to N, with N representing the discretization points in the temperature field. Additionally, considering the physical parameters would vary in time, the input $d_i^{(n)}$ in this problem is treated as a time

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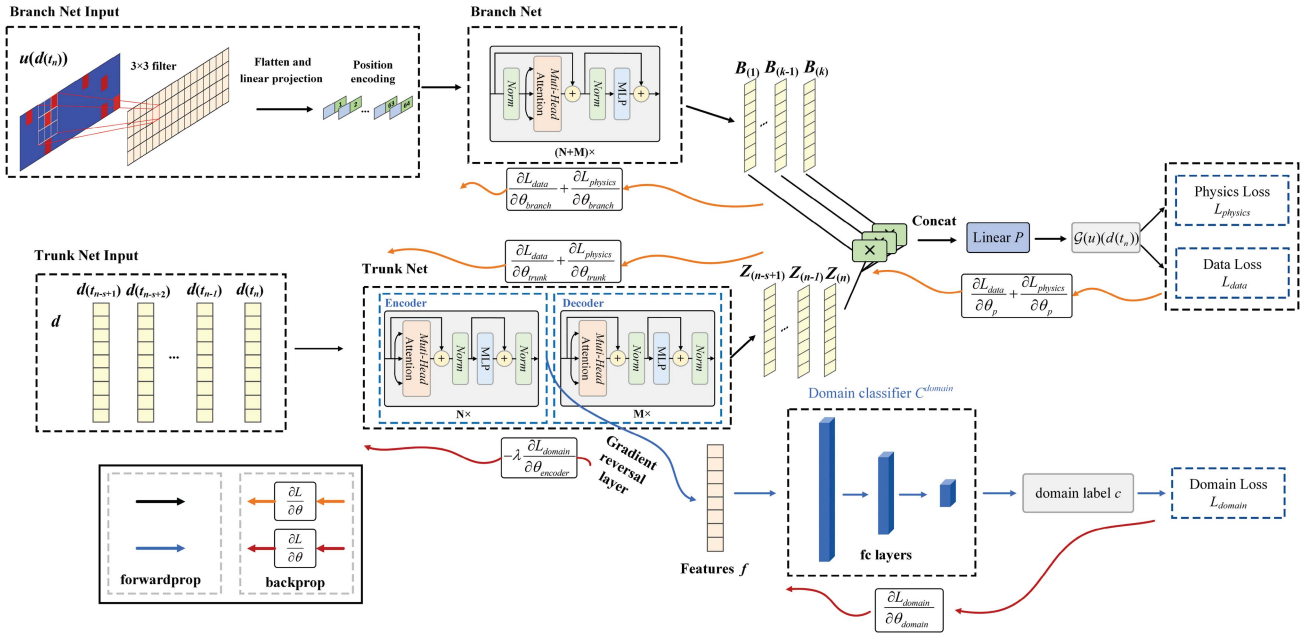


Fig. 2. Architecture of PIAN.

series, and historical data are used to capture temporal dynamic characteristics, which is shown in the following:

$$d_i^{(n)} = [d_i(t_{n-s+1}), \ldots, d_i(t_{n-1}), d_i(t_n)]^\tau. \tag{21}$$

To retain spatially distributed information, the nth time-step values from p monitoring sensors $[u(d_1^{(n)}), u(d_2^{(n)}), \ldots, u(d_p^{(n)})]^\tau$ is treated as a 1-channel image $u^{(n)} \in \mathbb{R}^{h\times w\times c}$ where h, w indicates the height and width of the image, c denotes the image channel, i.e., the sparse temperature value.

The architecture of the proposed PIAN is shown in Fig. 2. Initially, a "trunk" network accepts input d_i and produces hidden states Z. Additionally, we utilize a "branch" network to process image data $u^{(n)}$ and generate hidden states B. Finally, we merge these hidden states to obtain the corresponding temperature values $\mathcal{G}(u^{(n)})(d_i^{(n)})$. Our approach achieves the reconstruction of the temperature field by providing solutions for each coordinate point X_i located on the electrode surface. To leverage the computational efficiency and scalability advantages of multihead self-attention, we employ state-of-the-art backbone models, specifically the Transformer [21] and Vision Transformer [22], for the trunk and branch networks, respectively. The architecture of PIAN can also be seen as using different cognitive mechanisms to learn spatial and temporal concepts and to process sequential information through synchronization, which is similar to the mechanism by which the human brain inputs temporal information and contextual representations into the hippocampus using two different pathways [41]. The parameters of the entire neural network are denoted as θ. The parameters of the trunk net and branch net are represented by θ_{trunk} and θ_{branch}, respectively. The detailed forward pass is described as follows.

During the forward pass of the trunk net, the hidden states Z are obtained for input d through multiple transformer blocks (22), with h representing the hidden size of the neural networks. Each $Z_{(j)}$ represents the bias function of $d_i(t_j)$

$$Z = [Z_{(n-s+1)}, \ldots, Z_{(n-1)}, Z_{(n)}]^\tau \in \mathbb{R}^{s\times h}$$
$$Z_{(j)} = [z_1(d_i(t_j)), z_2(d_i(t_j)), \ldots, z_h(d_i(t_j))] \in \mathbb{R}^h. \tag{22}$$

In the forward pass of the branch net, the image $u^{(n)}$ is reshaped into a sequence of k patches Q using a 3×3 kernel filter to meet the input requirements of the Vision Transformer. This kernel size is appropriate when considering thermal dynamics, as each point energetically interacts with its surrounding points. Moreover, we introduce 1-D learnable position embeddings E_{pos} to retain spatial information

$$E_{\text{pos}} \in \mathbb{R}^{k\times h}, q_i \in \mathbb{R}^h$$
$$Q = [q_1, q_2, \ldots, q_k]^\tau + E_{\text{pos}} \in \mathbb{R}^{k\times h}. \tag{23}$$

Through branch net, the hidden states B of Q can be obtained (24), which represents the spatial dependent coefficient

$$B = [B_{(1)}, B_{(2)}, \ldots, B_{(k)}]^\tau \in \mathbb{R}^{k\times h}$$
$$B_{(j)} = [b_1(q_j), b_2(q_j), \ldots, b_h(q_j)] \in \mathbb{R}^h. \tag{24}$$

The spatial influence on $d_i(t_n)$ is represented by F function processing, denoted by H in the following:

$$H_j = F(B_{(j)}, Z_{(n)}) = \sum_{k=1}^{h} b_k(q_j) z_k(d_i(t_n))$$
$$H = [H_1, H_2, \ldots, H_k]^\tau \in \mathbb{R}^k. \tag{25}$$

Finally, we use a fully connected layer P on H to lift it back to the output space $\mathcal{V}$ and obtain $\mathcal{G}(u^{(n)})(d_i^{(n)})$. Efficient data enable neural networks to approximate the operator $\mathcal{G}$. However, ensuring performance becomes challenging in scenarios with limited data availability. To tackle this issue, we propose

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two effective mechanisms. The first mechanism is called the physics-informed mechanism, which allows for the embedding of domain knowledge into neural networks. The second mechanism is called the domain adversarial mechanism, which enables neural networks to extract domain-invariant features, ensuring the model performs well under unknown conditions.

1) Physics-Informed Mechanism: During the training process, our objective is to identify the optimal parameters of the neural network θ in order to approximate the operator. Simultaneously, we aim to ensure that the learned operator adhere to the physical laws, thereby enhancing the interpretability of the output and promoting its generalizability. Our learning object is shown in the following:

$$\begin{aligned}&\min_{\theta}\left(\sum_{j=s}^{M+s-1}\sum_{i=1}^{N}|\mathcal{G}_\theta(u^{(j)})(d_i^{(j)})-\overline{T}(t_j,d_i(t_j))|^2\right)\\&s.t.\ N\left[\mathcal{G}_\theta\left(u^{(j)}\right)\right]\left(d_i^{(j)}\right)=0\\&\quad B\left[\mathcal{G}_\theta\left(u^{(j)}\right)\right]\left(d_i^{(j)}\right)=0\end{aligned}\tag{26}$$

where $N[\mathcal{G}_\theta(u^{(j)})](d_i^{(j)})$ represents the PDE equation described in (1), while $B[\mathcal{G}_\theta(u^{(j)})](d_i^{(j)})$ encompasses boundary conditions formulated in (5).

The corresponding Lagrangian suggests an alternative approach of minimizing an unconstrained optimization objective (as shown in the following):

$$\min_{\theta} L_{\text{data}}(\theta)+L_{\text{physics}}(\theta).\tag{27}$$

$L_{\text{data}}(\theta)$ and $L_{\text{physics}}(\theta)$ represent deviations of model output from data measurements and physical laws, respectively. For the data loss function $L_{\text{data}}(\theta)$, we utilize the mean squared error (MSE) as depicted in the following:

$$L_{\text{data}}(\theta)=\frac{1}{MN}\sum_{j=s}^{M+s-1}\sum_{i=1}^{N}|\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)-\overline{T}\left(t_j,d_i(t_j)\right)|^2\tag{28}$$

where the ground truth temperature values are provided to supervise the learning. To account for the physics constraints, the physics loss $L_{\text{physics}}(\theta)$ incorporates residuals of the heat PDEs and boundary conditions. It consists of four parts: $L_{\text{PDE}}(\theta)$, $L_{\text{BC1}}(\theta)$, $L_{\text{BC2}}(\theta)$, and $L_{\text{BC3}}(\theta)$, illustrated as follows:

$$\begin{aligned}L_{\text{physics}}(\theta)=&\mu_1L_{\text{PDE}}(\theta)+\mu_2L_{\text{BC1}}(\theta)\\&+\mu_3L_{\text{BC2}}(\theta)+\mu_4L_{\text{BC3}}(\theta)\end{aligned}\tag{29}$$

$$\begin{aligned}L_{\text{PDE}}(\theta)=\frac{1}{MN}\sum_{j=s}^{M+s-1}\sum_{i=1}^{N}\Bigg(&\rho C_p\frac{\partial\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)}{\partial t}\\&-\lambda_x\frac{\partial^2G_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)}{\partial x^2}-\lambda_y\frac{\partial^2\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)}{\partial y^2}\\&-\lambda_z\frac{\partial^2\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)}{\partial z^2}-Q_{\text{heat}}\Bigg)^2\end{aligned}\tag{30}$$

$$\begin{aligned}L_{\text{BC1}}(\theta)=\frac{1}{MN}\sum_{j=s}^{M+s-1}\sum_{i=1}^{N}\Bigg[&h_{\text{conv1}}\left(\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)-T_{\text{amb}}\right)\\&-\lambda_x\frac{\partial\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)}{\partial x}\Bigg]^2\quad x=0,a\end{aligned}\tag{31}$$

$$\begin{aligned}L_{\text{BC2}}(\theta)=\frac{1}{MN}\sum_{j=s}^{M+s-1}\sum_{i=1}^{N}\Bigg[&h_{\text{conv2}}\left(\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)-T_{\text{liquid}}\right)\\&-\lambda_y\frac{\partial\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)}{\partial y}\Bigg]^2\quad y=0,b\end{aligned}\tag{32}$$

$$\begin{aligned}L_{\text{BC3}}(\theta)=\frac{1}{MN}\sum_{j=s}^{M+s-1}\sum_{i=1}^{N}\Bigg[&h_{\text{conv3}}\left(\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)-T_{\text{amb}}\right)\\&-\lambda_z\frac{\partial\mathcal{G}_\theta\left(u^{(j)}\right)\left(d_i^{(j)}\right)}{\partial z}\Bigg]^2\quad z=c\end{aligned}\tag{33}$$

where μ_1, μ_2, μ_3, and μ_4 are the corresponding coefficients. During implementation, automatic differentiation (AD) is employed to compute the derivative of the output with respect to the input. The entropic heat coefficient $\frac{dU_{\text{ocv}}}{dT}$ is determined as a function of SOC, following the open circuit potential (OCP) experiment outlined in [42]. The heat capacity C_p, thermal conductivity λ, and heat transfer coefficients h_{conv1}, h_{conv2}, and h_{conv3} are defined as learnable parameters. By utilizing surface temperature data from various locations during the optimization process, the converged parameters both satisfy the physical equations and minimize the disparity between the predicted and actual temperature values.

2) Domain Adversarial Mechanism: The challenging covariate shift problem occurs when the input distribution of training data and test data exist large difference, the model performance on the test set will inevitably decline. We define the input distribution as domain $\mathcal{M}$, which comprises two components: an input space $\mathcal{D}$ and a marginal probability distribution $P(D)$, where $D=\{d_i^{(j)}\}\in\mathcal{D}, i=1,\ldots,N, j=1,\ldots,M$. Specifically, the domains of the training set and the unknown operating condition are designated as $\mathcal{M}_s$ and $\mathcal{M}_t$, respectively, referred to as the source and target domains, $\mathcal{M}_s=\{\mathcal{D}_s,P(D_s)\}$, $\mathcal{M}_t=\{\mathcal{D}_t,P(D_t)\}$. In this study, the input space is the same, $\mathcal{D}_s=\mathcal{D}_t$, but the marginal probability distributions differ, $P(D_s)\neq P(D_t)$. Hence, the two domains, $\mathcal{M}_s$ and $\mathcal{M}_t$ are distinct: $\mathcal{M}_s\neq\mathcal{M}_t$. However, the relationship between the input space and the output space is governed by unchanging physical laws. Thus, we assume that there exists a new identical feature space that could generalize in both domains. In the real-world application, once the sparse sensors are determined, when encountering unknown operating conditions that were not included in the training set, it is impossible to collect the complete temperature field data. This problem can be

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viewed as an unsupervised domain adaptation (UDA) problems in transfer learning. Domain adversarial learning is one of the modern UDA methods minimize the distribution shift between training and testing domains. Based on the idea of a domain adversarial neural network (DANN) [43], we add a domain classifier to align feature representation of both domains.

During the training phase, we aim to extract domain-invariant features using labeled data from the source domain $(d_{i_s}^{(j)}, u_s^{(j)}, \overline{T}_s(d_i^{(j)}))$ and unlabeled data from the target domain $(d_{i_t}^{(j)}, u_t^{(j)})$. We decompose the trunk net into two parts: the encoder and decoder, parameterized by θ_{encoder} and θ_{decoder}. The first part contains N transformer blocks and is used to extract transferable features f from the input space D. The second part comprises M transformer blocks and maps the features f onto the hidden states Z of the trunk net. The feature f should be effectively utilized for the regression task while maintaining domain invariance. To achieve this, a domain classifier C^{domain} is added after the encoder E which is parameterized by θ_{domain}. This classifier accepts the feature representation f from encoder and output a label c to classify f according to which domain it belongs to ($c = 0$ if " f " is from the source domain, $c = 1$ if "f" is from the target domain).

The total loss of PIAN is demonstrated as follows:

$$L_{\text{total}}(\theta, \theta_{\text{domain}}) = L_{\text{data}}(\theta) + L_{\text{physics}}(\theta) - \beta L_{\text{domain}}(\theta_{\text{encoder}}, \theta_{\text{domain}}) \tag{34}$$

$$L_{\text{domain}}(\theta_{\text{encoder}}, \theta_{\text{domain}}) = -\frac{1}{P}\left(\sum_{k=1}^{P} c^k \log C_{\theta_{\text{domain}}}^{\text{domain}}\left(E_{\theta_{\text{encoder}}}\left(d^{(k)}\right)\right) + \sum_{k=1}^{P}\left(1-c^k\right)\log\left(1 - C_{\theta_{\text{domain}}}^{\text{domain}}\left(E_{\theta_{\text{encoder}}}\left(d^{(k)}\right)\right)\right)\right) \tag{35}$$

where $L_{\text{domain}}(\theta_{\text{encoder}}, \theta_{\text{domain}})$ represents the domain loss, formulated by the binary cross-entropy loss function (35). The tradeoff coefficient β controls the magnitude of the domain loss compared with other objectives. During training, the parameters θ_{encoder} of the feature mapping are optimized to maximize the L_{domain} loss (making the two feature distributions as similar as possible), while the parameters θ_{domain} of the domain classifier are optimized to minimize the L_{domain} loss (distinguish domain labels of the two feature distributions as much as possible). To implement this adversarial mechanism, a gradient reversal layer is inserted between encoder and the domain classifier, allowing end-to-end training of the method. Algorithm 1 illustrates the optimization strategy employed in the proposed deep transfer operator learning methods.

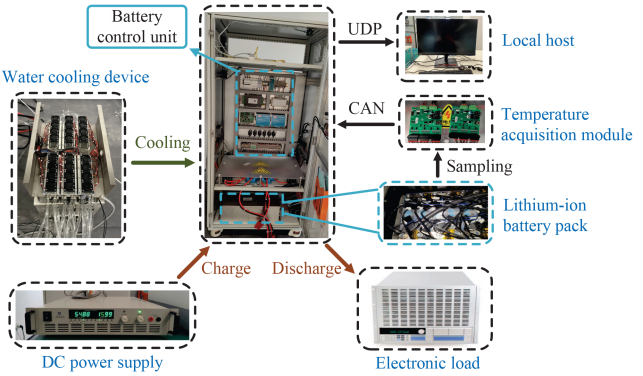


Fig. 3. Experimental setup of the energy storage platform.

Algorithm 1: Optimization of PIAN.

Require: source data: $\mathcal{S} = \{(d_{i_s}^{(j)}, u_s^{(j)}, \overline{T}_s(d_i^{(j)}))\}, i = 1, \ldots, N_s, j = 1, \ldots, M_s$,
target data:
$\mathcal{T} = \{(d_{i_t}^{(j)}, u_t^{(j)})\}, i = 1, \ldots, N_t, j = 1, \ldots, M_t$,
hyperparameters: μ_1, μ_2, μ_3, μ_4, β
Procedure:
Randomly initialize the parameters θ of the PIAN, domain classifier parameters θ_{domain}, and the learnable parameters C_p, λ, h_{conv1}, h_{conv2}, h_{conv3}.
for each epoch **do**
  for each batch **do**
    Forward $\mathcal{S}$ and compute $\mathcal{G}_\theta(u_s^{(j)})(d_{i_s}^{(j)})$ and feature f_s
    Forward $\mathcal{T}$ and compute $\mathcal{G}_\theta(u_t^{(j)})(d_{i_t}^{(j)})$ and feature f_t
    Compute source data loss L_{data} using (28)
    Compute source PDE loss L_{PDE} using (30)
    if source data point (x_s, y_s, z_s) is on the boundary **then**
      Compute boundary condition loss L_{BC} using (31)–(33)
      Compute physics loss L_{physics} using (29)
    else
      $L_{\text{physics}} = L_{\text{PDE}}$
    end if
    Compute domain loss L_{domain} using (35)
    Back-propagate with total loss L_{total} (34)
    Update network parameters θ, θ_{domain}, C_p, λ, h_{conv1}, h_{conv2}, h_{conv3}
  end for
end for

V. Experiment

A. Experiment Set up

The energy storage experimental platform is established, where the detail set up is depicted in Fig. 3. The energy storage system comprises a direct current (dc) power supply, a water-cooling device, a battery control unit, a local host, a temperature acquisition module, an electronic load, and LiB packs. The essential parameters of the chosen battery are listed in Table I. The dc power supply charges the battery while the electronic load discharges it. The water cooling device controls the flow of cooling liquid through the battery module on both sides at a certain flow rate using pulsewidth modulation (PWM) inverter pumps.

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TABLE I
PARAMETERS OF COMMERCIAL PRISMATIC LFP BATTERY

Name	Unit	Value
Cathode	/	LiFePO4
Anode	/	Graphite
Dimension(length×width×height)	mm3	148.3×26.7×91.5
Nominal capacity	Ah	32
Nominal voltage	V	3.20
Maximum cutoff voltage	V	3.65
Minimum cutoff voltage	V	2.50
Mass	g	735±50
Operating temperature range	°C	-20 to 55

B. Dataset

To validate our method, we collect data from the experimental platform. Temperature sensors are evenly distributed on a 4×16 grid of discrete battery electrode surfaces to sample temperature information. The battery control unit collects voltage, current, battery cell temperature, ambient temperature, and water temperature information. These measurements are transmitted to the local host via the Ethernet protocol. Additionally, we calculate the SOC of each cell using the ampere-hour counting method with a known initial SOC [44]. The relationship between OCV and SOC is obtained through a hybrid pulse power characterization (HPPC) test. All the aforementioned data are stored in a database for training algorithms. Specifically, the data are normalized and structured into triplets in the form of $[u, d, \mathcal{G}(u)(d)]$, as described in the following, for the purpose of training:

$$[u, d, \mathcal{G}(u)(d)] = \left[\begin{bmatrix} \vdots \\ u^{(j)} \\ u^{(j)} \\ \vdots \\ u^{(j)} \\ \vdots \end{bmatrix}, \begin{bmatrix} \vdots \\ d_1^{(j)} \\ d_2^{(j)} \\ \vdots \\ d_N^{(j)} \\ \vdots \end{bmatrix}, \begin{bmatrix} \vdots \\ \mathcal{G}\left(u^{(j)}\right)\left(d_1^{(j)}\right) \\ \mathcal{G}\left(u^{(j)}\right)\left(d_2^{(j)}\right) \\ \vdots \\ \mathcal{G}\left(u^{(j)}\right)\left(d_N^{(j)}\right) \\ \vdots \end{bmatrix} \right]. \tag{36}$$

The dataset comprises six distinct energy storage operating conditions with varying battery charge and discharge ratios. Each condition consists of 15 cycles of charging and discharging processes. The initial ambient temperature of each cycle is regulated by external air conditioning, fluctuating between 20 °C and 30 °C. This ensures that the initial conditions of each cycle are not identical, thus evaluating whether the algorithm can precisely reconstruct the temperature field under different initial conditions without retraining. From the available data, we randomly selected ten cycles as the training set, two cycles as the validation dataset to determine the architecture of neural networks, and three cycles as the test set.

VI. RESULTS

A. Sensor Placement

In this study, 80% of the collected training data under six operating conditions are used to train POD modes and QR sensors. The value of r is determined based on analyzing the decay of singular values and reconstruction error of the remaining 20% of the training data using coefficient $\widehat{a(t_j)}$. As shown in Fig. 4(a) and (b), we selected the value of r as the gradient changes significantly, resulting in $r = 8$ for this problem. After determining the number of sensors, we obtained the learned sensor position by performing QR factorization to matrix Θ, as depicted in Fig. 4(c).

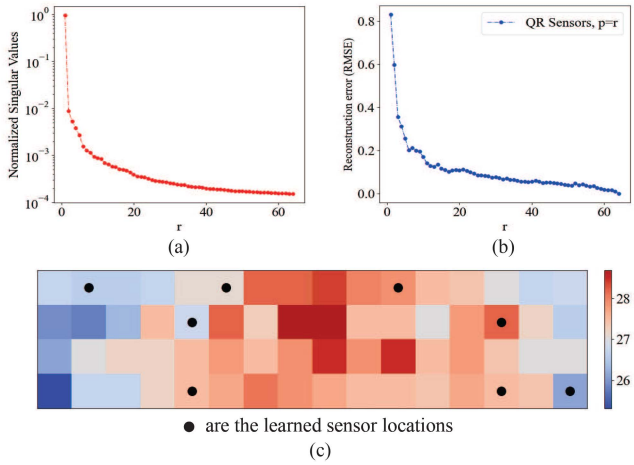


Fig. 4. (a) Normalized singular values with the rank r in tuncated SVD. (b) Reconstruction error with increasing number of QR sensors. (c) Learned sensor locations.

B. TFR via PIAN

PIAN demonstrates excellent performance in reconstructing battery temperature under different conditions and is capable of capturing uneven thermal behavior. Fig. 5(a) and (b) shows snapshots of the detected and predicted temperature fields in a test set of 0.5 C charging and discharging processes. Fig. 5(c) and (d) shows the reconstruction versus the measured results at the monitoring and other points. Our approach represents a mesh-free solution for tackling PDEs. It implies that, given any surface coordinates (x, y, z) of a battery module, physical parameters (L), and sparsely available temperature values (u) at monitoring points, our method can calculate the temperature for that particular point. Nevertheless, since the temperature at the monitoring point is used as input, neural networks can more readily learn to predict the temperature at the corresponding monitoring point than at other points. Therefore, we evaluated the temperature prediction errors for both the monitoring points and other locations. The results demonstrated that our method exhibited minimal temperature prediction errors at all points, demonstrating the generalization of our method.

1) Testing Time Complexity: The testing time is a crucial metric for evaluating the effectiveness of algorithms designed for real-time systems. After training, PIAN can directly reconstruct the temperature field based on the input. In this study, the parameter quantity for the optimal network structure of PIAN is 3.16 M, and the number of floating point operations (FLOPs) for each point solution is 117.02 M. PIAN can reconstruct temperature fields for a single time step within 0.048 s on an RTX 3090, demonstrating its suitability for real-time scenarios.

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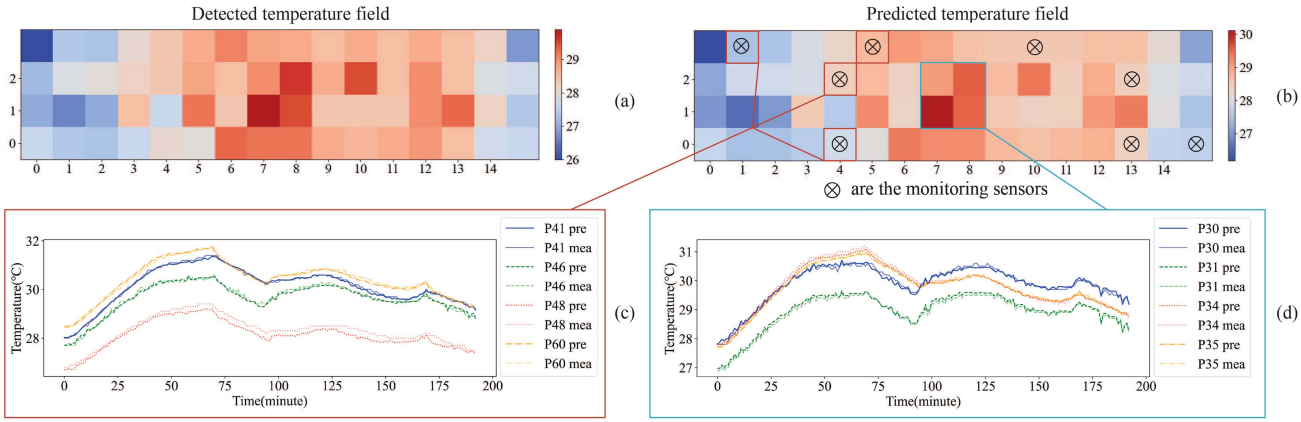


Fig. 5. (a) Snapshot of the detected temperature field measured by sensors. (b) Corresponding predicted temperature field by PIAN. (c) Monitoring points temperature reconstruction compared with measured values in a test cycle. (d) Other points of temperature reconstruction compared with measured values in a test cycle. "P41 mea" denotes the measured temperature at the 41st position in the discretized 64-grid temperature field. "P41 pre" represents the predicted temperature value at the 41st position.

TABLE II
PARAMETERS OF TRUNK NET AND BRANCH NET

Parameters	Module	
	Trunk net	Branch net
Number of attention heads	4	4
Value of N	2	2
Value of M	1	1
Hidden size	256	256
MLP size	512	512

2) Generalization Study: We conducted a task-based performance comparison to thoroughly assess the generalization capabilities of PIAN. For this purpose, a benchmark was established, and the performance of PIAN was evaluated alongside several state-of-the-art deep learning methods: MLP [45], UNet [46], and Vision Transformer [22]. MLP uses monitor sensor values as input to reconstruct the temperature field, aiming to learn the mapping from $\mathbb{R}^8$ to $\mathbb{R}^{64}$. UNet and Vision Transformer treat the input as a one-channel image, seeking to learn the mapping from $\mathbb{R}^{64}$ to $\mathbb{R}^{64}$. The mean absolute error (MAE) is applied to evaluate the reconstruction performance. The number of units in the hidden layers of all models is selected from the set $\{128, 256, 384, 512\}$. The number of heads for the multihead mechanism in the Vision Transformer and PIAN is chosen from the set $\{2, 4, 8, 16\}$. Regarding the determination of layers for all methods, the encoder and decoder layers of UNet demonstrate symmetrical increase or decrease. Similarly, the trunk network and branch network layers of PIAN also exhibit symmetrical changes. As for MLP and Vision Transformer, the number of layers is selected within the range of 2 to 6. Grid search is employed to determine the architecture of each method based on performance on the validation dataset. Fig. 6 illustrates the relationship between MAE performance and the number of layers for each method on the validation set under the 0.5 C charging and discharging condition.

The specific structure parameters of the trunk net and branch net in PIAN are available in Table II. The model was trained on 400-sized batches with Adam optimizer for 400 epochs at an initial learning rate of 0.0001, followed by a maximum of 300 epochs at a learning rate of 1e-5. The domain classifier consists of three fully connected layers, each with 1024 hidden states. The coefficients μ_1, μ_2, μ_3, μ_4, and β can be adaptively determined [47]. However, for the sake of easy implementation, these penalizing parameters are chosen from the set $\{0.01, 0.1, 1, 10\}$ and empirically set to 1, 0.1, 1, 1, and 1 after the grid searching process.

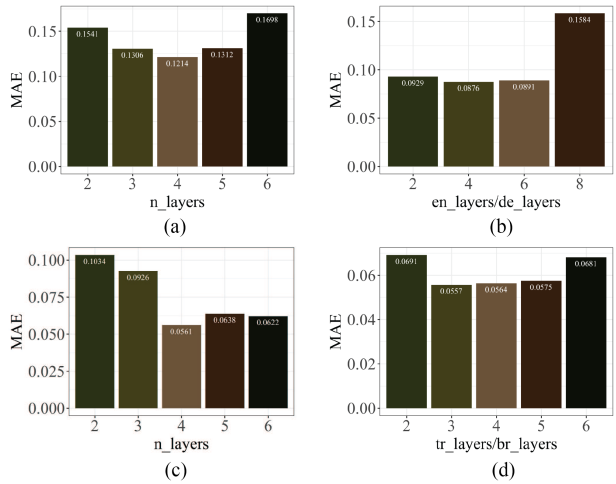


Fig. 6. Relationship between MAE performance and the network layers on the validation set for each method. (a) MLP. (b) UNet. (c) Vision Transformer. (d) PIAN. "n _ layers" means the number of layers. "en _ layers/de _ layers" denotes the number of layers of encoder/decoder. "tr _ layers/br _ layers" represents the number of layers in the trunk net/branch net.

We evaluate all the methods under the six test energy storage operating conditions. The training data and test data are under the same condition. All experiments run five times, and average results are presented in Table III. These results demonstrate that PIAN outperforms existing state-of-the-art deep learning methods in the TFR task.

To investigate cross-domain generalization, we defined battery charging and discharging at 0.5 C current as the source

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TABLE III
MAE OF DIFFERENT METHODS UNDER VARIOUS CONDITIONS

Condition	Method			
	MLP	UNet	ViT	PIAN(**Ours**)
0.5C charge 0.5C discharge[a]	0.187±0.002	0.134±0.015	0.119±0.004	**0.105**±0.008
0.5C charge 0.75C discharge	0.318±0.003	0.240±0.023	0.156±0.016	**0.123**±0.005
0.5C charge 1C discharge	0.148±0.013	0.105±0.012	0.157±0.007	**0.091**±0.003
1C charge 0.5C discharge	0.180±0.029	0.141±0.060	0.118±0.018	**0.103**±0.005
1C charge 0.75C discharge	0.211±0.013	0.177±0.057	0.125±0.009	**0.119**±0.002
1C charge 1C discharge	0.240±0.004	0.229±0.157	0.159±0.011	**0.093**±0.001

[a]"0.5C charge 0.5C discharge" means the battery is charged and discharged under a constant 0.5C current during operation.

The bold values indicate the best performance in terms of MAE for each condition, demonstrating the superior accuracy of our proposed method.

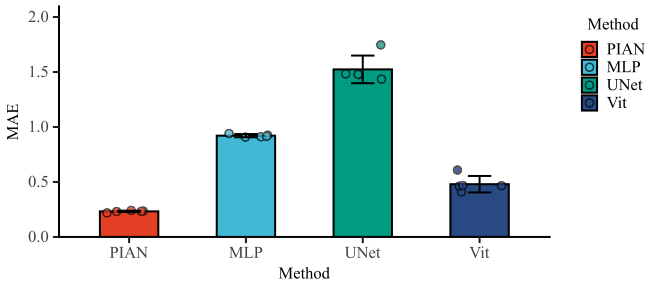


Fig. 7. Testing MAE error of PIAN and other methods on the target domain.

TABLE IV
MAE PERFORMANCE ON SOURCE AND TARGET TASK UNDER DIFFERENT DATA VOLUMES

Method	10% data		50% data		100% data	
	source	target	source	target	source	target
DeepONet[a]	0.488	0.828	0.370	0.720	0.139	0.319
DeepONet+ Physics	0.398	0.753	0.304	0.627	0.133	0.257
DeepONet+ Adaption	0.415	0.738	0.290	0.515	**0.094**	0.220
PIAN	**0.320**	**0.551**	**0.240**	**0.408**	0.097	**0.211**

[a]"DeepONet" in this table means the physics-informed and domain adversarial adaption weight of PIAN is set to 0.

The bold values represent the lowest MAE values, signifying the highest performance levels achieved by each method across different data volumes.

domain and battery charging and discharging at 1 C current as the target domain in this study. Since there exists a significant difference in heating rate and temperature inhomogeneity between these two working conditions, simulating the performance of the algorithm under unknown working conditions is rational. The results, shown in Fig. 7, demonstrate that PIAN outperforms other deep learning methods significantly in the target domain. Additionally, the standard deviation of MAE tested by the PIAN is smaller compared to other methods, where the standard deviation of MAE is larger and test error fluctuates greatly.

3) Ablation Study: We have found that physics-informed and domain adversarial adaptation mechanisms are effective, especially when limited data are available. To observe the improvement of the two mechanisms in the target domain, we gradually reduce the amount of data in the source domain. The results of the ablation experiment are presented in Table IV, indicating that both loss terms are effective in enhancing performance in the target domain.

VII. CONCLUSION

In this article, we proposed a deep transfer operator learning method named PIAN for TFR tasks in a LiB pack. Our results demonstrated strong generalization in real-world applications. We found that incorporating physics-informed and domain adversarial adaptation mechanisms enhanced performance when facing cases with limited data availability, which were known to be some of the most challenging cases for deep learning methods. In the future, we will explore the problem associated with optimal sensor placement, which has the potential to enhance the performance and robustness of the reconstruction methods.

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Yuchen Wang (Student Member, IEEE) received the B.S. degree in automation from North China Electric Power University, Beijing, China, in 2021. He is currently working toward the M.S. degree in control science and engineering with Shanghai Jiao Tong University, Shanghai, China.

His current research interests include physics-informed machine learning, neural symbolic regression, and neural operators, with a focus on their applications in control and power systems.



Can Xiong was born in Guizhou, China, in 1989. He received the B.S. degree in information engineering from Shanghai Jiao Tong University, Shanghai (SJTU), Shanghai, China, in 2013, and the M.S. degree in communication engineering from the University of Warwick, Coventry, U.K., in 2014, and the Ph.D. degree in electronic engineering from SJTU, in 2021.

After his Postdoctoral Research in automation from SJTU, in 2023, he is currently working with Technology Center Shanghai Electric Power Transmission and Distribution Group, Shanghai, China. His current research interest includes energy storage system and battery management system design, battery state estimation algorithm developing, compressed sensing, physics informed neural networks, phased antenna array diagnosis, and electromagnetic compatibility problems in high-speed circuit design.



Changjiang Ju was born in Anhui, China. He received the B.S. degree in automatic controlling from the Wuhan University of Technology, Wuhan, China, in 1998, and the M.S. and Ph.D. degrees in control science and engineering from Shanghai Jiao Tong University, Shanghai, China, in 2006 and 2019, respctively.

His research interests include smart energy, edge computing, and industrial process control.

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Genke Yang received the B.S. degree in mathematics from Shanxi University, Taiyuan, China, in 1984, the M.S. degree in mathematics from Xi'nan Normal University, Xi'an, China, in 1987, and the Ph.D. degree in systems engineering from Xi'an Jiaotong University, Xi'an, China, in 1998.

He has been a full-time Professor with the Department of Automation, Shanghai Jiao Tong University. He is currently a Member of the Collaborative Innovation Center for Advanced Ship and Deep-Sea Exploration, Shanghai. His research interests include supply chain management, logistics, production planning and scheduling, discrete event dynamics systems, and computer integrated manufacturing.



Yu-wang Chen received the Ph.D. degree in control theory and engineering from Shanghai Jiao Tong University, Shanghai, China, in 2008.

He is currently a Professor in Decision Sciences and Business Analytics with Alliance Manchester Business School, The University of Manchester (UoM), Manchester, U.K. Prior to joining UoM, he worked briefly as a Postdoctoral Research Fellow with the Department of Computer Science, Hong Kong Baptist University. His current research interests include decision and risk analysis under uncertainties, modeling and optimization of complex systems, operational research and data analytics.



Xiaotian Yu received the M.S. degree in control science and engineering from the Zhejiang University of Technology, Zhejiang, China, in 2020.

She is currently a Project Manager and Engineer specializing in energy storage safety with Ningbo Artificial Intelligence Research Institute, Shanghai Jiao Tong University, Shanghai, China. Her research interests include thermal management, life optimization, load forecasting, and safety assessment of energy storage systems.

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